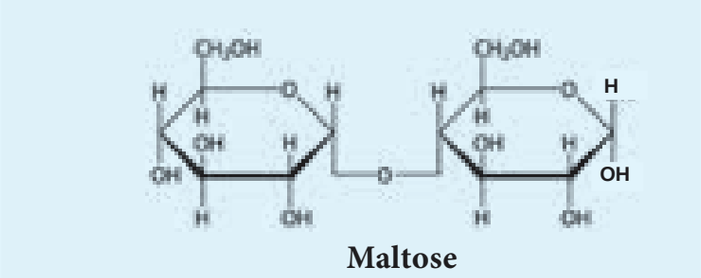
Maltose: Maltose derives its name from malt from which it is extracted. It is commonly called as malt sugar. Malt from sprouting barley is the major source of maltose. Maltose is produced during digestion of starch by the enzyme α-amylase.



Maltose

(α-D-glucopyranosyl-(1→4) α-D-glucopyranose)

Figure 14.11 Structure of Maltose

Maltose consists two molecules of α-D-glucose units linked by an α-1,4 glycosidic bond between anomeric carbon of one unit and C-4 of the other unit. Since one of the glucose has the carbonyl group intact, it also acts as a reducing sugar.

14.1.6 Polysaccharides:

Polysaccharides consist of large number of monosaccharide units bonded together by glycosidic bonds and are the most common form of carbohydrates. Since, they do not have sweet taste polysaccharides are called as non-sugars. They form linear and branched chain molecules.

Polysaccharides are classified into two types, namely, homopolysaccharides and heteropolysaccharides depending upon the constituent monosaccharides. Homopolysaccharides are composed of only one type of monosaccharides while the heteropolysaccharides are composed of more than one. Example: starch, cellulose and glycogen (homopolysaccharides); hyaluronic acid and heparin (heteropolysaccharides).

STARCH

Starch is used for energy storage in plants. Potatoes, corn, wheat and rice are the rich sources of starch. It is a polymer of glucose in which glucose molecules are lined by α(1,4) glycosidic bonds. Starch can be separated into two fractions namely, water soluble amylose and water insoluble amylopectin. Starch contains about 20 % of amylose and about 80% of amylocpectin.

Amylose is composed of unbranched chains upto 4000 α-D-glucose molecules joined by α(1,4)glycosidic bonds. Amylopetin contains chains upto 10000 α-D-glucose molecules linked by α(1,4)glycosidic bonds. In addition, there is a branching from linear chain. At branch points, new chains of 24 to 30 glucose molecules are linked by α(1,6)glycosidic bonds. With iodine solution amylose gives blue colour while amylopectin gives a purple colour.

XII U14-Biomolecules_New.indd 248 11-01-2021 16:37:46